

# Indexing & Data Sync Guide

---

## Overview

---

We have migrated to a **Snapshot Sync Architecture** to avoid iCloud database corruption.

- **Mac Mini (Indexer):** Indexes data locally, then pushes a snapshot to iCloud.
  - **MacBook (Consumer):** Pulls the snapshot from iCloud to local storage for querying.
  - **Safe & Fast:** Databases run on local SSDs. iCloud is only used for transfer.
- 

## Unified Procedures

---

### 1. Indexing (Run on Mac Mini)

**Script:** `Scripts/run_indexing.sh`

This unified script performs a complete re-index of your vault into all three databases:

1. **ChromaDB** (Vector Search) - Port 8000
2. **Kimi Graph** (Note Graph) - Port 8002
3. **LightRAG** (Entity Graph) - Port 8001

**Usage:**

```
# Full Re-Index
./Scripts/run_indexing.sh
```

- You will be prompted to confirm a full re-index (deletes old data).
  - **Time:** ~30-60 mins for 2k notes.
  - **Cost:** ~\$1.00 (Kimi/OpenRouter).
- 

### 2. Export Data (Run on Mac Mini)

**Script:** `Scripts/sync/push.sh`

After indexing is complete, push the fresh data to the iCloud staging area.

This creates a stable snapshot in `~/Library/Mobile Documents/com~apple~CloudDocs/ai/RAG/obsidian_rag/data/export_stage`.

**Usage:**

```
./Scripts/sync/push.sh
```

---

### 3. Import Data (Run on MacBook)

**Script:** `Scripts/sync/pull.sh`

To update your MacBook with the latest SOTA data from the Mini, run the pull script.

This stops your local services, pulls the data from iCloud to your local SSD (`~/obsidian_rag_local_data`), and restarts services.

**Usage:**

```
./Scripts/sync/pull.sh
```

---

### 1a. Targeted Indexing (Advanced)

If you only need to update **one** specific index without running the full pipeline:

#### Vector Only (ChromaDB)

- **Use when:** You added new notes and just want them searchable.
- **Command:**

```
python src/indexing/index_vault.py --refresh
```

#### NetworkX Graph Only

- **Use when:** You want to refresh wiki-links without re-embedding text.
- **Command:**

```
docker exec -it obsidian-graph-service python  
/app/src/services/kimi_graph_builder.py
```

#### LightRAG Only (Entities)

- **Use when:** You want to extract entities/relations for "Deep Thinking".
- **Command:**

```
./Scripts/index_with_lightrag.sh
```

---

### 4. Daily Usage (Run on Both)

**Script:** `Scripts/start_obsidian_rag.sh`

To start the system for querying/browsing.

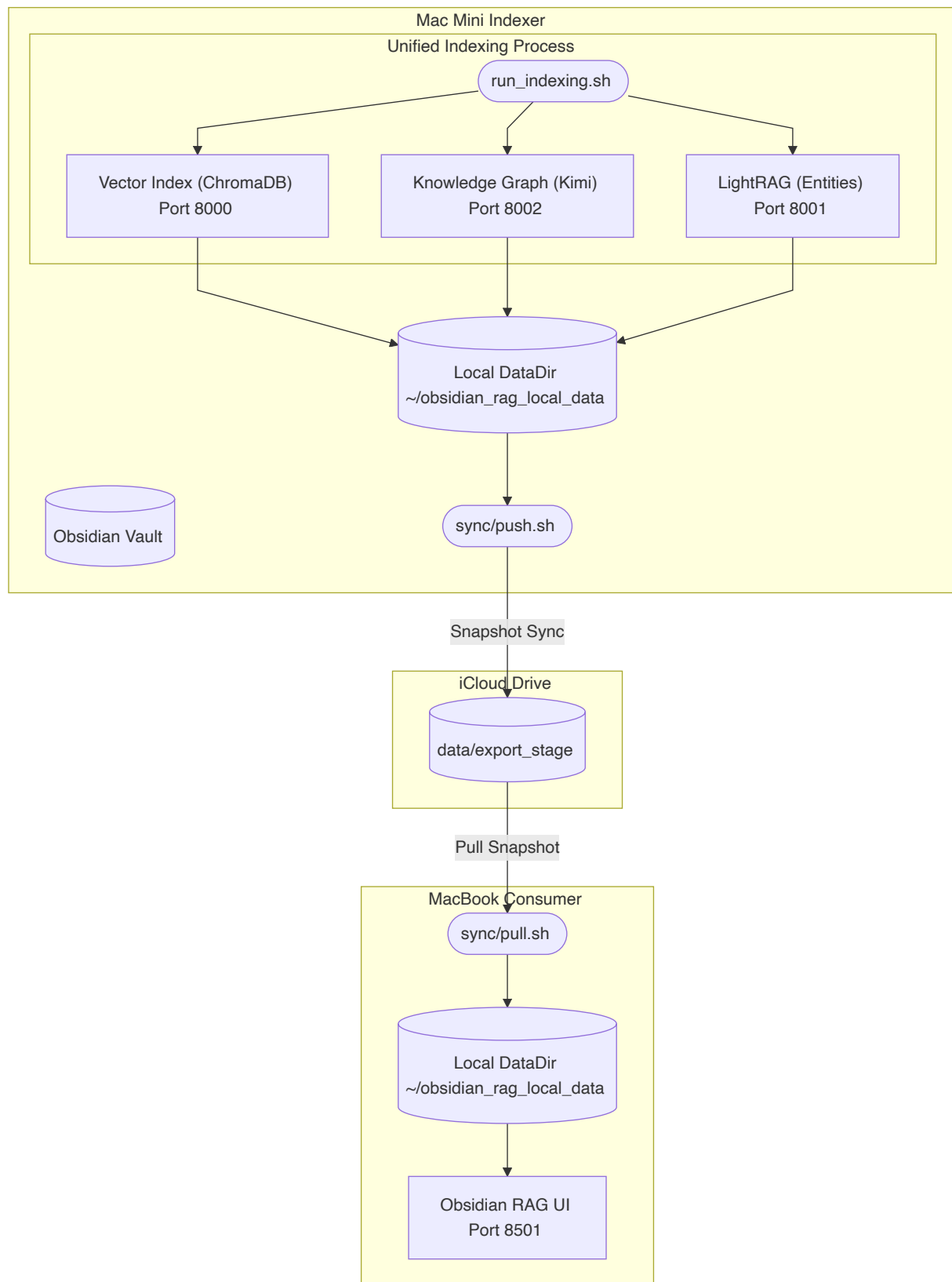
**Usage:**

```
# Start Docker Services
./Scripts/start_obsidian_rag.sh

# Stop Services
./Scripts/stop_obsidian_rag.sh
```

---

# Architecture Diagram



# Troubleshooting

---

## "Naive" Results on MacBook?

If you see "Naive" mode instead of "Hybrid":

1. **Check Sync:** Did you run `pull.sh` recently?
2. **Check Ollama:** Ensure Ollama is running (`ollama serve`). The embedding model (`nomic-embed-text`) must be available for vectors to work.
3. **Check Logs:**

```
docker logs obsidian-lightrag
```

Verify that incoming queries show `Requested mode: 'hybrid'` and `Effective mode: 'hybrid'`.

## "Database Locked" Errors?

- Should NOT happen with this new architecture.
- If it does, ensure you are NOT trying to index on the MacBook or access the `data/export_stage` directly from Docker. Always use the local copies.

---

**Last Updated:** January 25, 2026